

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0943

Sistrunk 100 kV
March 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0943. The Interconnection Customer proposes a 43.6 MW solar photovoltaic facility interconnecting at the Sistrunk 100 kV substation in the Illinois area. The request was received on February 17, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 263 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,067,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wallisville-Basscreek 100 kV	Stapley-Arlington Hts 100 kV	110.6	139	8.1
Mouton Substation-Treat 230 kV Line	El Paso-Ravenna 500 kV	105.9	575	5.7
Stapley-Prospect Hts 100 kV	Prospect Heights-Poter County 100 kV L	105.2	137	9.9
Medora-Hampton Rd 345 kV	Simpson Stuart Tap-Fishkill Plains 345 kV	104.1	1,928	6.7
Involta Substation-Chamberlin 230 kV line	Hylebos-Ravenna 500 kV Line	104.0	500	5.2
Eastown Rd-Swan Valley 230 kV	Swan Vly-Trego 500 kV Line	101.4	1,027	6.2
Mason St-Herald-Twin Cities 230 kV	New Departure-Mason St 230 kV L	101.0	936	5.4
Chamberlin-Treat 230 kV Line	Ravenna Substation-El Paso 500 kV L	100.7	633	5.7
Burris Pit-Hebo 230 kV	Trego-Beaver 500 kV	100.5	1,340	7.8
Twin County Ranch-Richland Substation 230 kV	Troy Energy Llc-Richland 500 kV L	100.4	1,454	19.2
Burris Pit-James De Young 230 kV	Trego-Beaver 500 kV L	100.3	1,365	7.8
Goose Haven Substation-Rosemore B 100 kV Cir. 1	Goose Haven-Rosemore B 100 kV	100.1	215	17.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Basscreek-Wallisville 100 kV	Rebuild line with higher-capacity conductor	\$4,518,000	100% (\$4,518,000)
Mouton-Treat 230 kV L	Reconductor line	\$9,863,000	0% (\$34,000)
Stapley-Prospect Hts 100 kV	Reconductor line	\$791,000	26% (\$205,000)
Medora-Hampton Rd 345 kV line	Reconductor line	\$7,170,000	0% (\$24,000)
Chamberlin-Involta 230 kV	Reconductor line	\$6,027,000	0% (\$20,000)
Eastown Rd-Swan Valley 230 kV	Reconductor line	\$3,230,000	0% (\$11,000)
Mason St-Herald-Twin Cities 230 kV	Reconductor line	\$15,130,000	0% (\$52,000)
Treat-Chamberlin 230 kV	Reconductor line	\$5,671,000	0% (\$19,000)
Burris Pit-Hebo 230 kV	Reconductor line	\$9,456,000	0% (\$32,000)
Twin County Ranch-Richland 230 kV L	Reconductor line	\$17,956,000	0% (\$62,000)
Burris Pit-James De Young 230 kV	Reconductor line	\$16,534,000	0% (\$56,000)
Rosemore B-Goos Haven Substation 100 kV	Reconductor line	\$313,000	11% (\$34,000)
	Total allocated to project		\$5,067,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.